

HW2

STAT 651

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1: BDA3 2.20

a)

$p(y|\theta) \sim \text{Exp}(\theta)$, $\pi(\theta) \sim \text{Gamma}(\alpha, \beta)$. We observe that $y \geq 100$. What is $p(\theta|y \geq 100)$?

$$p(\theta|y \geq 100) \propto p(y \geq 100|\theta)\pi(\theta) \propto [1 - (1 - e^{-100\theta})]\theta^{\alpha-1}e^{-\beta\theta} = e^{-100\theta}\theta^{\alpha-1}e^{-\beta\theta} = \theta^{\alpha-1}e^{-\theta(\beta+100)}$$

This is a Gamma distribution: $p(\theta|y \geq 100) \propto \text{Gamma}(\alpha, \beta + 100)$

We can use the mean and variance of the Gamma to give us:

Posterior Mean: $\frac{\alpha}{\beta+100}$

Posterior Variance: $\frac{\alpha}{(\beta+100)^2}$

b)

$$p(\theta|y = 100) \propto \theta e^{-100\theta}\theta^{\alpha-1}e^{-\beta\theta} = \theta^{\alpha}e^{-\theta(\beta+100)}$$

This is a Gamma distribution: $p(\theta|y = 100) \propto \text{Gamma}(\alpha + 1, \beta + 100)$

We can use the mean and variance of the Gamma to give us:

Posterior Mean: $\frac{\alpha+1}{\beta+100}$

Posterior Variance: $\frac{\alpha+1}{(\beta+100)^2}$

c)

On average the intuition is correct, but it is not guaranteed that for all values of y , knowing the exact value will certainly reduce the variance. (2.8) only states that on average, more knowledge will reduce variance. In this case, the value of 100 is one such value that actually ends up increasing the variance for whatever reason.

2: BDA 3.3

$$n_c = 32, \bar{y}_c = 1.013, s_c = .24, n_t = 36, \bar{y}_t = 1.173, s_t = .2$$

a)

Let $p(y_c|\mu_c) = N(\mu_c, \sigma_c^2); \pi(\mu_c), \pi(\mu_t), \pi(\ln(\sigma_c)), \pi(\ln(\sigma_t)) = Unif(0, 1)$

$p(\mu_c|y_c) = p(y_c|\mu_c)\pi(\mu_c)$ This is a normal distribution with an uninformative uniform prior. This is shown in the book to be $t_{n-1}(\bar{y}, s^2/n)$:

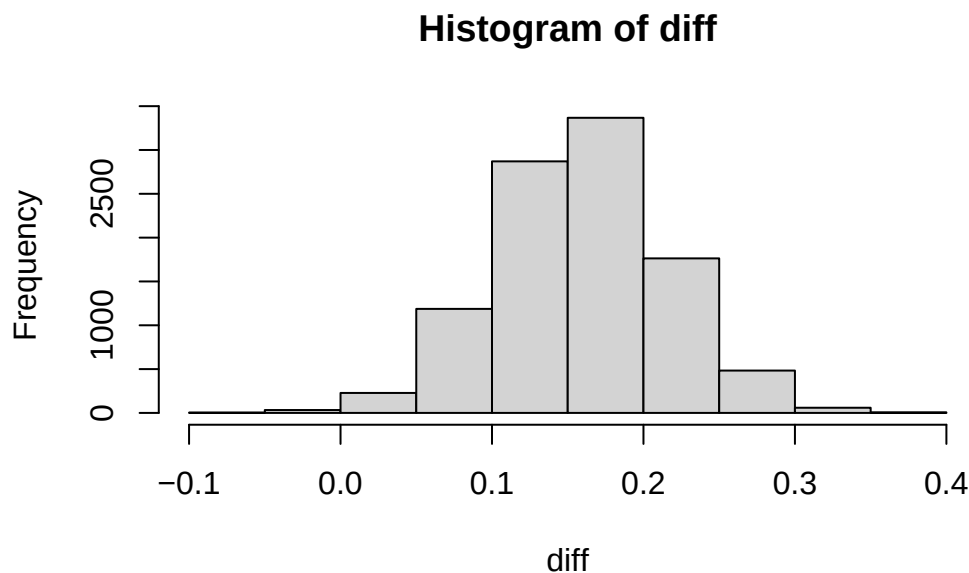
$$\Rightarrow p(\mu_c|y_c) = t_{n_c-1}(\bar{y}_c, s_c^2/n_c) = t_{31}(1.013, .24^2/32)$$

Similarly we can find:

$$\Rightarrow p(\mu_t|y_t) = t_{n_t-1}(\bar{y}_t, s_t^2/n_t) = t_{35}(1.173, .2^2/36)$$

b)

```
c <- rt(10000, 31)*.24/sqrt(32) + 1.013
t <- rt(10000, 35)*.2/sqrt(36) + 1.173
diff <- t-c
hist(diff)
```



```
lower <- round(quantile(diff, .025), 4)
upper <- round(quantile(diff, .975), 4)
```

95% interval: 0.0487 to 0.2715

3

WHEN GENERATING FINAL PDF, REMOVE EVAL:FALSE, and uncomment/move results out of code

I'm using one of the priors from the HW2 key, the moderately informative prior: $\alpha = 11, \beta = 5500$

Below is my STAN code:

```
data {
  int<lower=0> N;
  array[N] int y;
}

parameters {
  real theta;
}

model {
  y ~ exponential(theta);
  theta ~ gamma(11, 5500); //From HW2 Key
}
```

```
set.seed(1337)
rivers <- as.numeric(readLines("rivers.dat"))

suppressPackageStartupMessages(library("rstan"))

data_stan <- list(N = length(rivers), y = rivers)

params <- c("theta")

n_cores <- parallel::detectCores() # Count how many cores are available
options(mc.cores = n_cores)

fit_stan <- stan(model_code = readLines("rivers.stan"),
  data = data_stan, pars = params,
  iter = 5000, warmup = 500, thin = 5, chains = n_cores)
```

Warning in readLines("rivers.stan"): incomplete final line found on
'rivers.stan'

```
summary(fit_stan)

mode <- function(x) {
  ux <- unique(x)
```

```

  ux[which.max(tabulate(match(x, ux)))]
}

sim_stan <- extract(fit_stan) # collect samples from all chains
post_thetas <- sim_stan$theta
post_mean <- mean(post_thetas)
post_med <- median(post_thetas)
post_mode <- mode(post_thetas)
post_var <- var(post_thetas)
post_sd <- sd(post_thetas)
post_int <- c(quantile(post_thetas, .025), quantile(post_thetas, .975))

```

We get the following:

Posterior Mean: 0.0017121

Posterior Median: 0.0017091

Posterior Mode: 0.0015136

Posterior Variance: 1.896238×10^{-8}

Posterior Standard Deviation: 1.3770396×10^{-4}

95% Credibel Interval: 0.0014533, 0.0019887

4

$$x_t | t_i, \theta \stackrel{ind}{\sim} Pois(\lambda_i), \lambda_i = t_i \theta$$

a)

$$L(\theta | t_i | x_i) = \prod \frac{\lambda_i^{x_i} e^{-\lambda_i}}{x_i!} = e^{\ln[\prod \frac{\lambda_i^{x_i} e^{-\lambda_i}}{x_i!}]} = e^{\sum [x_i(\ln(\lambda_i)) - \lambda_i - \ln(x_i!)]} = e^{\sum x_i \ln(\lambda_i) - \sum \lambda_i - \sum \ln(x_i!)} = e^{\sum x_i (\ln(\theta) + \ln(t_i)) - \theta \sum t_i - \sum \ln(x_i!)} = e^{\sum x_i \ln(t_i) - \sum \ln(x_i!)} e^{\sum x_i \ln(\theta) - \theta \sum t_i}$$

b)

$$\eta(\theta) = \ln(\theta)$$

$$T(x_i) = \sum x_i$$

c)

$$Gamma(\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} = \frac{\beta^\alpha}{\Gamma(\alpha)} e^{(\alpha-1)\ln(\theta) - \beta\theta}$$

d)

Expert expects no more than 10 failures in a month, meaning the failure rate in a day should be well below 1/3 (in a typical month of 30 days).

I used Google to look up parameters for a Gamma distribution that would put the upper 95% interval at 1/3 and got the following: $\alpha = 2, \beta = .0703$

e)

$$p(\theta|x_i) \propto e^{\ln(\theta) \sum x_i - \theta \sum t_i} e^{(\alpha-1)\ln(\theta) - \beta\theta} = e^{\ln(\theta)(\sum x_i + \alpha - 1) - \theta(\beta + \sum t_i)}$$

This is the Natural Parameter form of the Gamma with $\alpha = \sum x_i + \alpha$ and $\beta = \beta + \sum t_i$

$$\Rightarrow p(\theta|x_i) \sim \text{Gamma}(\alpha + \sum x_i, \beta + \sum t_i)$$

f & g)

My Stan Code:

```
data {  
  int<lower=0> N;  
  array[N] int x;  
  array[N] int t;  
  real alpha;  
  real beta;  
}  
  
parameters {  
  real theta;  
}  
  
model {  
  real sum_x = sum(x);  
  real sum_t = sum(t);  
  x ~ exponential(theta);  
  theta ~ gamma(alpha, beta);  
}
```

```
comps <- read.table("computer-failures-process.dat", header = TRUE)  
  
alpha = 2  
beta = .0703  
  
data_stan <- list(N = length(comps$x), x = comps$x, t = comps$t, alpha = alpha, beta = beta)  
  
params <- c("theta")
```

```
n_cores <- parallel::detectCores() # Count how many cores are available
options(mc.cores = n_cores)

fit_stan <- stan(model_code = readLines("comps.stan"),
               data = data_stan, pars = params,
               iter = 5000, warmup = 500, thin = 5, chains = n_cores)
```

Warning in readLines("comps.stan"): incomplete final line found on 'comps.stan'

```
sim_stan <- extract(fit_stan) # collect samples from all chains
post_thetas <- sim_stan$theta
post_int_4 <- c(quantile(post_thetas, .025), quantile(post_thetas, .975))
index_hpd <- which(sim_stan$lp__ >= quantile(sim_stan$lp__, .05))
hpd_int <- c(min(sim_stan$theta[index_hpd]), max(sim_stan$theta[index_hpd]))
```

We get the following:

95% Equal Tailed: 0.292846, 0.769976

HPD Interval (95%): 0.2753503, 0.7385895

h)

```
prior_prob <- pgamma(.1, shape = alpha, rate = beta)
post_prob <- pgamma(.1, shape = (alpha + sum(comps$x)), rate = (beta + sum(comps$t)))
```

Prior: $p(\theta < .1 | x_i) = \text{pgamma}(.1, \text{shape} = 2, \text{rate} = .0703) \implies p(H_0) = 2.4594945 \times 10^{-5}, p(H_1) = 0.9999754$

Posterior: $p(\theta < .1 | x_i) = \text{pgamma}(.1, \text{shape} = 2, \text{rate} = .0703) \implies p(H_0 | \text{data}) = 0.7814108, p(H_1 | \text{data}) = 0.2185892$

Bayes Factor: $\frac{p(H_1 | \text{data}) / p(H_1)}{p(H_0 | \text{data}) / p(H_0)} = 6.8802738 \times 10^{-6}$

This means that even though the prior was for H_1 , there was a significant amount of evidence in the data to heavily swing it more against H_1 in favor of H_0

i)

The following finds the probability for equals, as well as for the epsilon ranges. For my generated samples, it appears that none of them are in the appointed ranges, due to my minimum parameter value being 0.1548978.

```
mean(post_thetas == .1)
```

```
[1] 0
```

```
epsilon <- c(.005, .001, .0001)
lapply(epsilon, \(e) cat('Epsilon:', e, 'Probability: ', mean((post_thetas >= .1 - e) & (post_t
```

```
Epsilon: 0.005 Probability: 0
Epsilon: 0.001 Probability: 0
Epsilon: 1e-04 Probability: 0
```

```
[[1]]
NULL
```

```
[[2]]
NULL
```

```
[[3]]
NULL
```

```
min(post_thetas)
```

```
[1] 0.1548978
```

For the posteriors, no, it doesn't really make sense to look at probability of equaling a value because they are draws, and thus dependent on at least one draw equalling that exact value. Intervals make much more sense.

5

$$f(x_i|\theta) = \theta^{x_i}(1-\theta)^{1-x_i}$$

$$\phi = \ln\left(\frac{\theta}{1-\theta}\right)$$

a)

$$\phi = \ln\left(\frac{\theta}{1-\theta}\right) \implies e^\phi = \frac{\theta}{1-\theta} \implies e^\phi - \theta e^\phi = \theta \implies e^\phi = \theta(1 + e^\phi) \implies \theta = \frac{e^\phi}{1+e^\phi}$$

$$f(x_i|\theta) = \theta^{x_i}(1-\theta)^{1-x_i} \implies f(x_i|\phi) = \left[\frac{e^\phi}{1+e^\phi}\right]^{x_i} \left[1 - \frac{e^\phi}{1+e^\phi}\right]^{1-x_i}$$

$$\text{When } x_i = 1 \implies f(x_i|\phi) = \frac{e^\phi}{1+e^\phi}$$

$$\text{When } x_i = 0 \implies f(x_i|\phi) = 1 - \frac{e^\phi}{1+e^\phi} = \frac{1+e^\phi}{1+e^\phi} - \frac{e^\phi}{1+e^\phi} = \frac{1}{1+e^\phi}$$

The only thing that differs between the two is the numerator, which is e^ϕ when $x = 1$, and 1 when $x_1 = 0$. This can be represented by just $e^{\phi x_i}$

$$\implies \underline{f(x_i|\phi) = \frac{e^{\phi x_i}}{1+e^\phi} I\{x_i \in \{0, 1\}\}}$$

b)

$$f(x|\phi) = \prod \frac{e^{\phi x_i}}{1+e^\phi} = \frac{e^{\phi \sum x_i}}{(1+e^\phi)^n}$$

$$l(x|\phi) = \phi \sum x_i - \ln(1+e^\phi)$$

$$\frac{d}{d\phi} l(x|\phi) = \sum x_i - \frac{e^\phi}{1+e^\phi}$$

$$I = \mathbb{E}[(\frac{dl}{d\phi})^2] = \mathbb{E}[-\frac{e^\phi}{(1+e^\phi)^2}] = \frac{e^\phi}{(1+e^\phi)^2}$$

$$\pi_\phi^1(\phi) = \sqrt{I(\phi)} = \sqrt{\frac{e^\phi}{(1+e^\phi)^2}} = \frac{e^{\phi/2}}{1+e^\phi}$$

c)

$$\begin{aligned} \pi_\theta^1(\theta) \left| \frac{\partial \theta}{\partial \phi} \right| &= \text{Beta}(1/2, 1/2) \frac{\partial}{\partial \phi} \frac{e^\phi}{1+e^\phi} \\ &\propto \theta^{-1/2} (1-\theta)^{-1/2} \frac{e^\phi}{(1+e^\phi)^2} \\ &= \left[\frac{e^\phi}{1+e^\phi} \right]^{-1/2} \left(1 - \frac{e^\phi}{1+e^\phi} \right)^{-1/2} \frac{e^\phi}{(1+e^\phi)^2} \\ &= \frac{e^{\phi-1/2\phi}}{(1+e^\phi)^{2-1/2-1/2}} \\ &= \frac{e^{\phi/2}}{(1+e^\phi)} \end{aligned}$$

That is the same as $\pi_\phi^1(\phi)$ shown in part (b).

6

$$y_i|\mu, \tau \sim N(\mu, 1/\tau), \text{Var}(y_i|\mu, \tau) = 1/\tau$$

$$\pi(\mu, \tau) \propto 1/\tau$$

a)

Per (3.2) in BDA3, we have that the Joint Posterior is (just replacing σ^2 with $1/\tau$):

$$p(\mu, \tau|y) \propto \tau^{(n-1)/2} \exp(-\tau/2 [\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2])$$

To get the marginal for τ we integrate out μ :

$$\begin{aligned}
p(\tau|y) &\propto \int_{\mu} \tau^{(n-1)/2} \exp(-\tau/(2\tau) [\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2]) \\
&= \tau^{(n-1)/2} \exp(-\tau/2 \sum_{i=1}^n (y_i - \bar{y})^2) \int_{\mu} \exp(-\tau/2n(\bar{y} - \mu)^2) \text{the red bit is a normal kernel} \\
&\propto \tau^{(n-1)/2} \exp(-\tau/2 \sum_{i=1}^n (y_i - \bar{y})^2) \tau^{1/2} \\
&= \tau^{(n-1)/2-1} \exp(-\tau/2 \sum_{i=1}^n (y_i - \bar{y})^2)
\end{aligned}$$

This is proportional to a Chi-Squared distribution: χ_{n-1}^2

b)

$t = \sqrt{n}(\mu - \bar{y})/s$ where $s^2 = \sum (y_i - \bar{y})^2 / (n-1)$

Show $t \propto (1 + \frac{t^2}{n-1})^{-n/2}$.

In BDA3, we have that the marginal posterior for μ is: $p(\mu|y) \propto \left[1 + \frac{n(\mu - \bar{y})^2}{(n-1)s^2}\right]^{n/2} =$
 $\left[1 + \left(\frac{\sqrt{n}(\mu - \bar{y})}{s}\right)^2 \frac{1}{n-1}\right]^{n/2}$

Plugging in t : $p(t|y) \propto \left[1 + \frac{t^2}{n-1}\right]^{-n/2}$

This is a t-distribution: t_{n-1}

I'm not too sure what to comment on, but very interesting that a reparameterization that seems a little crazy would end up resulting in the t-distribution.

Acknowledgements

Estimated time

I estimate this assignment took me in total around 11 hours to complete

Resources Used

I used the book quite a lot, class slides. I used Google AI to help me understand how to derive a Jeffrey's prior, and how to setup STAN stuff